PROJECT OUTLINE

Centralized Coal Logistics & Penalty Control System

What We Are Going to Do

Currently, at **Pakur, Dumka, and Kurwa railway sidings**, operational data is maintained in **multiple Excel sheets**.
Under this project, **the same data will simply be entered once into a centralized application database** instead of Excel.

Data entry remains the same — only the medium changes from Excel to application.

After that:

- All calculations
- Penalty checks
- Analysis
- Dashboards
  will be **automatically managed by the system**.

Why This Change Is Required

- Today, the **same data is used at multiple levels** (operations, finance, management)
- Manual handling and reuse of Excel data leads to:
  - Errors
  - Delays
  - Inconsistent reporting
- Since the **data source is one**, but utilization is different, **manual processing increases risk**

This system removes that risk by **using one centralized data source**.

Main Objective

1. **Minimize railway penalties**
2. **Provide real-time, consolidated visibility**
3. **Enable proactive control instead of post-billing analysis**

Key Business Benefit

- Today, penalties are known **after RR is received**
- With this system:
  - We can **predict penalties before RR**
  - Take **corrective action in advance**

Especially for **overloading**, which is the main penalty contributor.

How Penalties Will Be Reduced

- Loader-wise and wagon-wise loading data will be analysed automatically

- If overloading trend is detected at:
    - A particular siding
    - A particular loader
    - A particular rake
      → the system will generate alerts

This allows **timely correction**, not repeated mistakes.

What Management Will See

Real-time dashboards showing:

- Siding-wise (Pakur / Dumka / Kurwa) penalties
- Rake-wise and date-wise dispatch
- Power plant–wise performance
- Loader-wise overloading trends

All **visible at a glance** for planning and strategy.

Implementation Approach

- Phase-wise, low-risk rollout
- No operational disruption
- Excel replacement with centralized system
- RR remains the final financial authority

One-Line Note

"To implement a centralized coal logistics system that replaces Excel-based data handling with a single application, enabling real-time visibility, penalty prevention, and data-driven decision-making across Pakur, Dumka, and Kurwa sidings"

End-to-End Work Flow (System-Driven)

Coal Receipt at Railway Siding

- Coal received shift-wise by trucks
- Operator enters:
  - Trips received
  - Quantity received
- System updates **live siding stock**

Indent Raising for Rake

- When stock becomes rake-sufficient:
  - Siding In-Charge raises **indent in system**
- System records:
  - Indent time
  - Target rake quantity

Rake Placement by Railways

- Railway places rake (immediate or delayed)
- **Exact rake placement time captured**
- System automatically:
  - Starts time tracking
  - Flags placement delays (for analysis)

TXR Inspection & Unfit Wagon Identification

- TXR inspection carried out by Railways
- Unfit wagons identified and marked
- Operator enters:
  - Unfit wagon numbers
  - Time taken for TXR activity
- System logs:
  - TXR duration
  - Fit vs unfit wagons

Loading Time Starts (Critical Control Point)

- Loading time starts **from rake placement**
- System automatically:
  - Starts **3-hour free time counter**
  - Shows live time balance
- Any delay beyond free time:
  - **Auto-flags penalty risk**

Wagon-Wise Loading (Most Important Stage)

- Loaders load wagons one by one
- Loader data captured via:

    - Loader application **or**
    - Manual log book entry (as done today)

  Data captured:

- Wagon number
- Loader ID
- Quantity loaded

  Same data as today — only digitized

In-Motion Weighment

- Rake moved to in-motion weighment track
- Weighment slip received from Railways
- Operator enters / uploads:
    - In-motion weighment data

Automatic Comparison & Early Warning

- System compares:
    - Loader-noted wagon weight
    - In-motion weighment weight
- If deviation or overload trend found:
    - System generates **alert**
    - Loader / supervisor informed immediately

  This happens **before RR**, not after billing

Rake Movement Monitoring

- System captures:
    - Time rake moved after loading
- If rake movement is delayed:
    - **Alert generated to speed up process**
- Helps avoid:
    - Demurrage
    - Secondary delays

RR as Final Authority

- Commercial RR received later
- RR data uploaded
- System:
    - Matches RR penalties with predicted penalties
    - Confirms savings / deviations